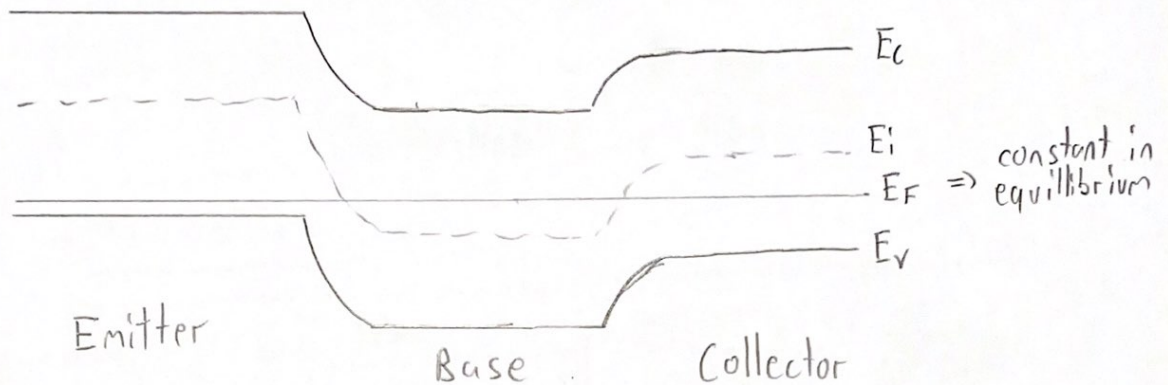
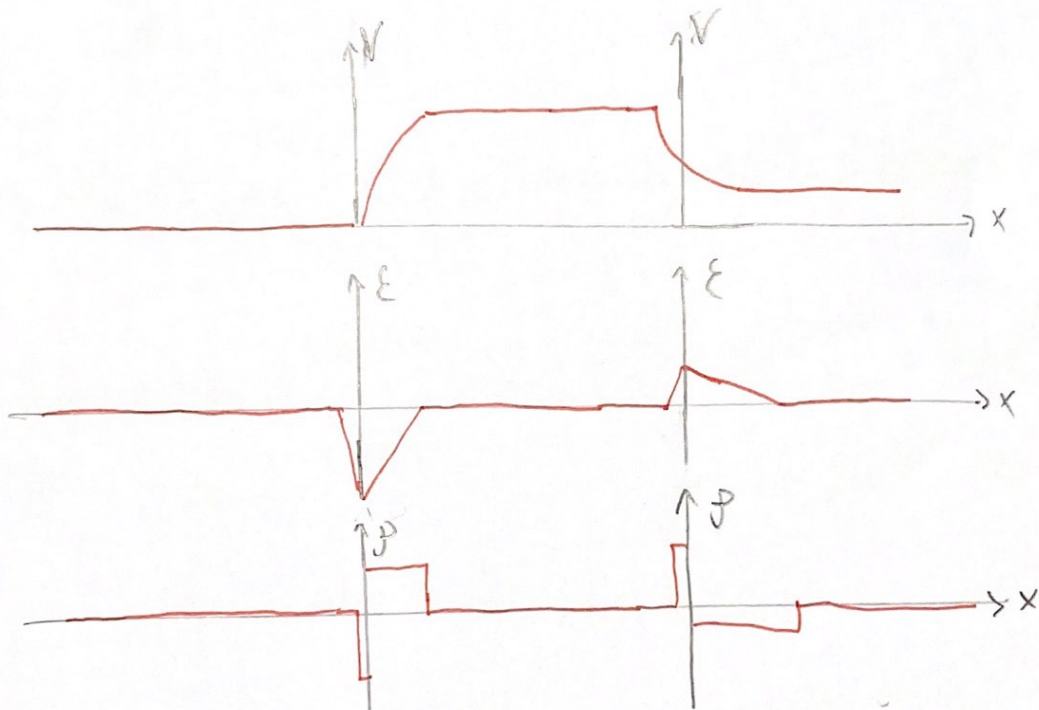


1)

a)



b)



$$c) V_{EC} = V_E - V_C = -\frac{1}{q} \underbrace{(E_{c-\text{emitter}} - E_{c-\text{collector}})}_{\text{conduction band energies}} = -\frac{1}{q} (E_{i-\text{emitter}} - E_{i-\text{collector}})$$

On emitter side

$$E_i - E_F = kT \ln\left(\frac{p_0}{n_i}\right) = 0.026 \text{ eV} \ln\left(\frac{5 \times 10^{18}}{1.5 \times 10^{10}}\right) = 0.51 \text{ eV}$$

On collector side

$$E_i - E_F = 0.026 \text{ eV} \ln\left(\frac{10^{15}}{1.5 \times 10^{10}}\right) = 0.29 \text{ eV}$$

$$V_{EC} = -\frac{1}{q} (0.51 \text{ eV} - 0.29 \text{ eV}) = -0.22 \text{ V}$$

★ Since E_F is constant throughout the BJT, we can find the difference between E_i 's on each side

d) We can think of the BJT as 2 separate PN junctions

$$\xi_{\max-EB} = -\frac{qN_{DB}}{\epsilon_s} x_{n0-EB} = \frac{qN_{AE}}{\epsilon_s} x_{p0-EB} \Rightarrow \text{For EB junction}$$

$$\xi_{\max-CB} = \frac{qN_{DB}}{\epsilon_s} x_{n0-CB} = \frac{qN_{AE}}{\epsilon_s} x_{p0-CB} \Rightarrow \text{For CB junction}$$

For EB junction

$$x_{n0} = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \left(\frac{N_A}{N_D(N_A + N_D)} \right)}$$

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) = 0.026 \text{ eV} \ln \left(\frac{5 \times 10^{18}}{2.25 \times 10^{20}} \right)$$

$$= 0.86 \text{ eV}$$

$$= \sqrt{\frac{2 \times 11.8 \times 8.85 \times 10^{-14} \times 0.86}{1.6 \times 10^{-19}} \left(\frac{5 \times 10^{18}}{10^{16} (5.01 \times 10^{18})} \right)}$$

$$= 3.35 \times 10^{-5} \text{ cm}$$

$$\xi_{\max-EB} = -\frac{1.6 \times 10^{-19} \times 10^{16}}{11.8 \times 8.85 \times 10^{-14}} 3.35 \times 10^{-5} = -51320 \text{ V/cm}$$

FOR CB junction

$$x_{n0} = \sqrt{\frac{2 \times 11.8 \times 8.85 \times 10^{-14} \times 0.64}{1.6 \times 10^{-19}} \frac{10^{15}}{10^{16} \times 11 \times 10^{15}}}$$

$$= 8.71 \times 10^{-6} \text{ cm}$$

$$V_{bi} = 0.026 \text{ eV} \ln \left(\frac{10^{37}}{2.25 \times 10^{20}} \right) = 0.64 \text{ eV}$$

$$\xi_{\max-CB} = \frac{1.6 \times 10^{-19} \times 10^{16}}{11.8 \times 8.85 \times 10^{-14}} 8.71 \times 10^{-6} = 13344 \text{ V/cm}$$

e) Again we can find depletion width by the PN junction assumption as:

$$X_{n-EB} = \sqrt{\frac{2\epsilon_s (V_{bi} - V_A)}{q} \left(\frac{N_{AE}}{N_{DB}(N_{AE} + N_{DB})} \right)} = \sqrt{\frac{2 \times 1.04 \times 10^{-12} \times 0.16}{1.6 \times 10^{-19}} \times \frac{5 \times 10^{18}}{10^{16} (5 \times 10^{18} + 10^{16})}}$$

$$= 1.44 \times 10^{-5} \text{ cm}$$

$$X_{n-CB} = \sqrt{\frac{2\epsilon_s (V_{bi} + V_A)}{q} \left(\frac{N_{AC}}{N_{DB}(N_{AC} + N_{DB})} \right)} = \sqrt{\frac{2 \times 1.04 \times 10^{-12} \times 1.64}{1.6 \times 10^{-19}} \times \frac{10^{15}}{10^{16} \times 11 \times 10^{15}}}$$

$$= 1.4 \times 10^{-5} \text{ cm}$$

$$W = W_b - X_{n-EB} - X_{n-CB} = 1 \mu\text{m} - 0.144 \mu\text{m} - 0.14 \mu\text{m} = 0.716 \mu\text{m}$$

f) $\Delta p_B(0) = p_{B0} (e^{qV_{EB}/kT} - 1)$ with $p_{B0} = \frac{n_i^2}{N_{DB}} = \frac{2.25 \times 10^{20}}{10^{16}} = 2.25 \times 10^4$

$\Delta p_B(W) = p_{B0} (e^{qV_{CB}/kT} - 1)$ $V_{EB} = 0.7 \text{ V}$ $V_{CB} = -1 \text{ V}$

$$\Delta p_B(0) = 2.25 \times 10^4 \text{ cm}^{-3} (e^{0.7/0.026} - 1) = 1.1 \times 10^{16} \text{ cm}^{-3}$$

$$\Delta p_B(W) = 2.25 \times 10^4 \text{ cm}^{-3} (e^{\underset{\approx 0}{-1/0.026}} - 1) = -2.25 \times 10^4 \text{ cm}^{-3} \Rightarrow p_B(W) = 0$$

g) $\alpha_{dc} = \frac{1}{1 + \frac{n_i^2}{n_B^2} \frac{D_E}{D_B} \frac{N_B}{N_E} \frac{W}{L_E} + \frac{1}{2} \left(\frac{W}{L_B} \right)^2}$

$$D_E = \frac{kT}{q} \mu_n = 0.026 \text{ V} \times 150 \text{ cm}^2/\text{Vs} = 3.9 \text{ cm}^2/\text{s}$$

$$D_B = \frac{kT}{q} \mu_p = 0.026 \text{ V} \times 400 \text{ cm}^2/\text{Vs} = 10.4 \text{ cm}^2/\text{s}$$

$$L_E = \sqrt{D_E \tau_E} = \sqrt{10^{-17} \times 3.9} = 1.97 \times 10^{-5} \text{ cm}$$

$$L_B = \sqrt{D_B \tau_B} = \sqrt{2.5 \times 10^{-9} \times 10.4} = 1.61 \times 10^{-4} \text{ cm}$$

$$n_{iE} = n_{iB}$$

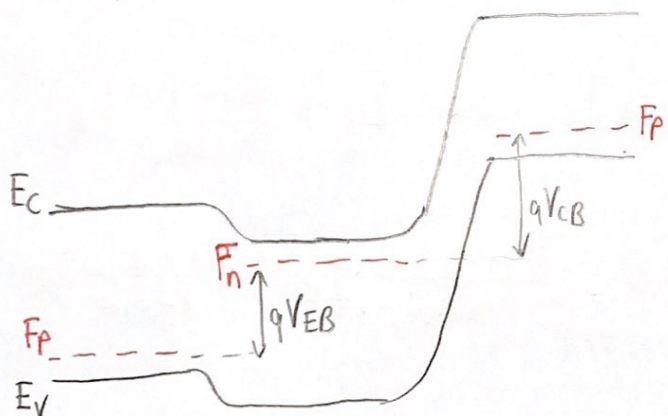
→ same material, temperature

$$\alpha_{dc} = \frac{1}{1 + \frac{3.9}{10.4} \times \frac{10^{16}}{5 \times 10^{18}} \times \frac{0.716 \times 10^{-4}}{1.97 \times 10^{-5}} + \frac{1}{2} \left(\frac{0.716 \times 10^{-4}}{1.61 \times 10^{-4}} \right)^2} = 0.91$$

$$\beta_{dc} = \frac{\alpha_{dc}}{1 - \alpha_{dc}} = \frac{0.91}{0.09} = 10.1$$

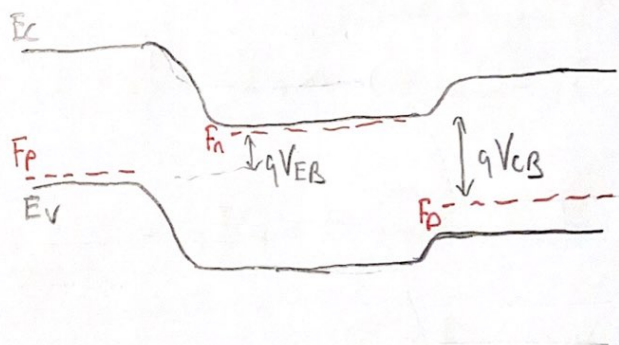
2. Forward Active

a) EB forward biased - CB reverse biased

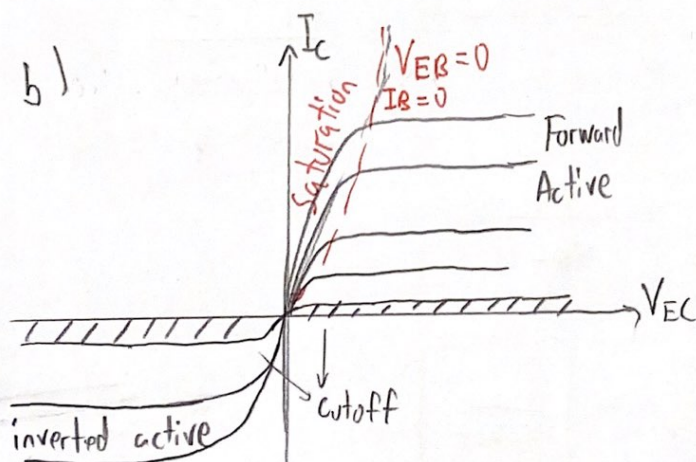


Saturation

Both EB, CB forward biased



b)



c) If τ_B decreases
 L_B decreases

$$\alpha_T = \frac{1}{1 + \frac{1}{2} \left(\frac{W}{L_B} \right)^2} \quad \text{decreases}$$

$$\gamma = \frac{1}{1 + \frac{D_E}{D_B} \frac{N_B}{N_E} \frac{W}{L_E}} \quad \text{no change}$$

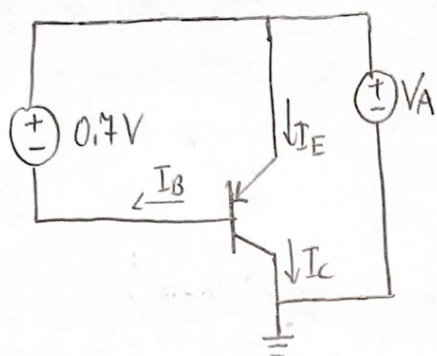
$$\beta_{dc} = \frac{1}{(---) + \frac{1}{2} \left(\frac{W}{L_B} \right)^2} \quad \text{decreases}$$

$$I_C = \alpha_F I_{F0} e^{qV_{BE}/kT} + I_{R0} (1 - e^{q(V_{EC} - V_{BE})/kT})$$

* This is expected since recombination in the base drastically increases

★

3)



$$V_{EB} = 0.7V$$

$$\alpha_F = 0.98, \alpha_R = 0.7$$

$$I_{FO} = 10^{-13} A, I_{RO} = 7.14 \times 10^{-14} A$$

a) $V_{EC} = 0.05$ $V_{CB} = V_{EB} - V_{EC} = 0.65 V \Rightarrow$ CB junction forward biased

* Since both EB and CB junctions are forward biased
BJT is in saturation mode.

$$I_C = \alpha_F I_{FO} (e^{qV_{EB}/kT} - 1) - I_{RO} (e^{qV_{CB}/kT} - 1)$$

$$= 0.98 \times 10^{-13} (e^{0.7/0.026} - 1) - 7.14 \times 10^{-14} (e^{0.65/0.026} - 1) = 0.043 A$$

$$I_E = I_{FO} (e^{qV_{EB}/kT} - 1) - \alpha_R I_{RO} (e^{qV_{CB}/kT} - 1)$$

$$= 10^{-13} (e^{0.7/0.026} - 1) - 0.7 \times 7.14 \times 10^{-14} (e^{0.65/0.026} - 1) = 0.0457 A$$

$$I_B = I_E - I_C = 2.7 mA$$

b) $V_{EB} = 0.7, V_{EC} = 0.7 \Rightarrow V_{CB} = 0 V \Rightarrow$ Between active and saturation modes

$$I_E = 0.98 \times 10^{-13} (e^{0.7/0.026} - 1) = 0.04828 A$$

$$I_E = 10^{-13} (e^{0.7/0.026} - 1) = 0.04927 A$$

$$I_B = 0.999 mA$$

c) $V_{EB} = 0.7V$, $V_{EC} = 2V \Rightarrow V_{CB} = -1.3V \Rightarrow \text{Active Mode}$

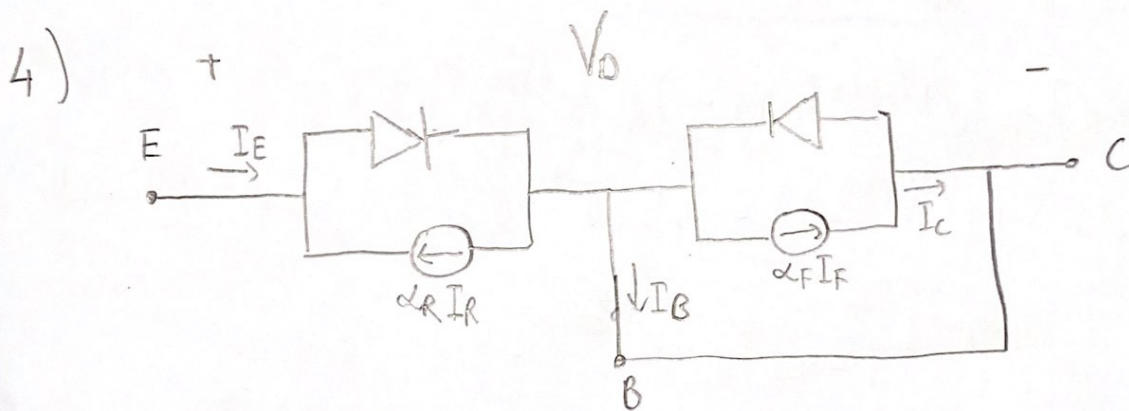
$$I_C = 0.98 \times 10^{-13} (e^{0.7/0.026} - 1) - 4.14 \times 10^{-14} (e^{\underbrace{-1.3/0.026}_{\approx 0}} - 1)$$

$$= 0.04828 A$$

$$I_E = 10^{-13} (e^{0.7/0.026} - 1) - 0.7 \times 4.14 \times 10^{-14} (e^{\underbrace{-1.3/0.026}_{\approx 0}} - 1)$$

$$= 0.04927 A$$

$$I_B \approx 0.999 \text{ mA}$$



$$V_D = V_{EC} = V_{EB} \Rightarrow V_{CB} = 0 \Rightarrow \text{short circuited}$$

$$I_E = I_D \quad I_D = I_{F0} (e^{qV_D/kT} - 1) //$$

5)

$$I_{CEO} \Rightarrow I_C \text{ when } I_B = 0$$

$$I_{BEO} \Rightarrow I_C \text{ when } I_E = 0$$

$$I_C = \alpha_F I_{F0} (e^{qV_{EB}/kT} - 1) - I_{R0} (e^{qV_{CB}/kT} - 1)$$

$$I_E = I_{F0} (e^{qV_{EB}/kT} - 1) - \alpha_R I_{R0} (e^{qV_{CB}/kT} - 1)$$

$$\text{Let } I_F = I_{F0} (e^{qV_{EB}/kT} - 1) \text{ and } I_R = I_{R0} (e^{qV_{CB}/kT} - 1)$$

When $I_B = 0$

$$I_C = I_E \Rightarrow \alpha_F I_F - I_R = I_F - \alpha_R I_R \Rightarrow I_R = \frac{\alpha_F - 1}{1 - \alpha_R} I_F$$

$$I_{CEO} = I_C = \alpha_F I_E - \frac{\alpha_F - 1}{1 - \alpha_R} I_F = \frac{1 - \alpha_F \alpha_R}{1 - \alpha_R} I_F = \frac{\alpha_F \alpha_R - 1}{1 - \alpha_F} I_R$$

When $I_E = 0$

$$\alpha_R I_R = I_F \Rightarrow I_{BEO} = I_C = \alpha_F I_F + \frac{I_F}{\alpha_R} = \frac{\alpha_F \alpha_R - 1}{\alpha_R} I_F = (\alpha_F \alpha_R - 1) I_R$$

$$\frac{I_{CEO}}{I_{CBO}} = \frac{1 - \alpha_F \alpha_R}{1 - \alpha_R} I_F \frac{\alpha_R}{\alpha_F \alpha_R - 1} \frac{1}{I_F} = \frac{\alpha_R}{\alpha_R - 1}$$

$$\text{OR}$$

$$= \frac{\alpha_F \alpha_R - 1}{1 - \alpha_F} I_R \frac{1}{\alpha_F \alpha_R - 1} \frac{1}{I_R} = \boxed{\frac{1}{1 - \alpha_F}}$$

* When there is no base current holes do not recombine with electrons in the base so all the current coming from the emitter directly passes onto collector.
 When there is no emitter current only the base current will be passed onto collector which is small due to self-constructed reverse bias